Midterm Test
F18CE – Multivariable Calculus
Test 1 2025/26

11 February 2026

Duration: 30 minutes

No calculators allowed

1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function defined by

$$f(x,y) = \begin{cases} \frac{kx^2}{x-y}, & x \neq y \\ 0, & x = y \end{cases},$$

where k is a fixed real number.

(a) Use the definition to evaluate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(0,0)$. **[4]**

(b) Determine whether f is continuous at $(0,0)$. **[5]**

(c) Hence, determine whether f is differentiable at $(0,0)$. **[1]**

2. Let $g : \mathbb{R}^2 \to \mathbb{R}$ be a function defined as

$$g(x,y) = e^{x^2+y^2}.$$

(a) Evaluate the gradient of g at point $(0,1)$. **[1]**

(b) Write the equation of the tangent plane to the surface $z = g(x,y)$ at $(0,1)$. **[2]**

(c) Let $v = (v_1, v_2)$ be a unit vector. **[4]**

(i) Find the directional derivative of g at $(0,1)$ in the direction of v.

(ii) Find v so that the directional derivative in (i) vanishes.

End of paper

ANSWER SCHEME

1. (a) $\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{kh^2}{h-0}-0}{h} = k$ (2 marks)

$\frac{\partial f}{\partial y}(0,0) = \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{0}{0-h}}{h} = 0\,.$ (2 marks)

(b) When $k = 0$: Then $f(x,y) = 0$ for all $(x,y) \in \mathbb{R}^2$.

Then obviously f is continuous and differentiable at (0,0). (1 mark)

When $k \neq 0$: Consider pathways $y = x \pm x^2$ (1 mark)

$\lim_{(x,y)\to(0,0)} f(x, x+x^2) = \lim_{x\to 0} \frac{kx^2}{x-x-x^2} = -k,$ (1 mark)

$\lim_{(x,y)\to(0,0)} f(x, x-x^2) = \lim_{x\to 0} \frac{kx^2}{x-x+x^2} = k \neq -k \ (\because k \neq 0)$ (1 mark)

Thus, $\lim_{(x,y)\to(0,0)} f(x,y) \nexists$. So, $f(x,y)$ is not continuous at $(0,0)$ when $k \neq 0$. (1 mark)

Alternative: Consider pathway $y = x - x^3$, then $\lim_{(x,y)\to(0,0)} f(x, x-x^3) = \lim_{x\to 0} \frac{k}{x} \nexists$.

(c) When $k = 0$, f is differentiable at (0,0). When $k \neq 0$, f is not differentiable at (0,0) since f is not continuous at (0,0). (1 mark)

2. $g(x,y) = e^{x^2+y^2}$.

(a) $\nabla g = \left(2xe^{x^2+y^2}, 2ye^{x^2+y^2}\right) \Rightarrow \nabla g(0,1) = (0,2e)$ (1 mark)

(b) $z = e + 0 + 2e(y-1)$ (2 marks)

(c) (i) $\partial_v g(0,1) = (0,2e)\cdot(v_1, v_2) = 2ev_2$ (2 marks)

(ii) $\partial_v g(0,1) = 0 \Rightarrow v_2 = 0\,.$ (1 mark)

So, $v_1 = \pm 1$ as v is a unit vector. Hence, $v = (\pm 1, 0)$. (1 mark)